



INTRODUCTION

Colostrum is the first source of nutrition and immunity for calves, supporting passive transfer of immunoglobulins (Ig) and early growth. However, yield and quality **vary** widely among dairy cows, and the **factors** associated with this **variability** are not fully understood.

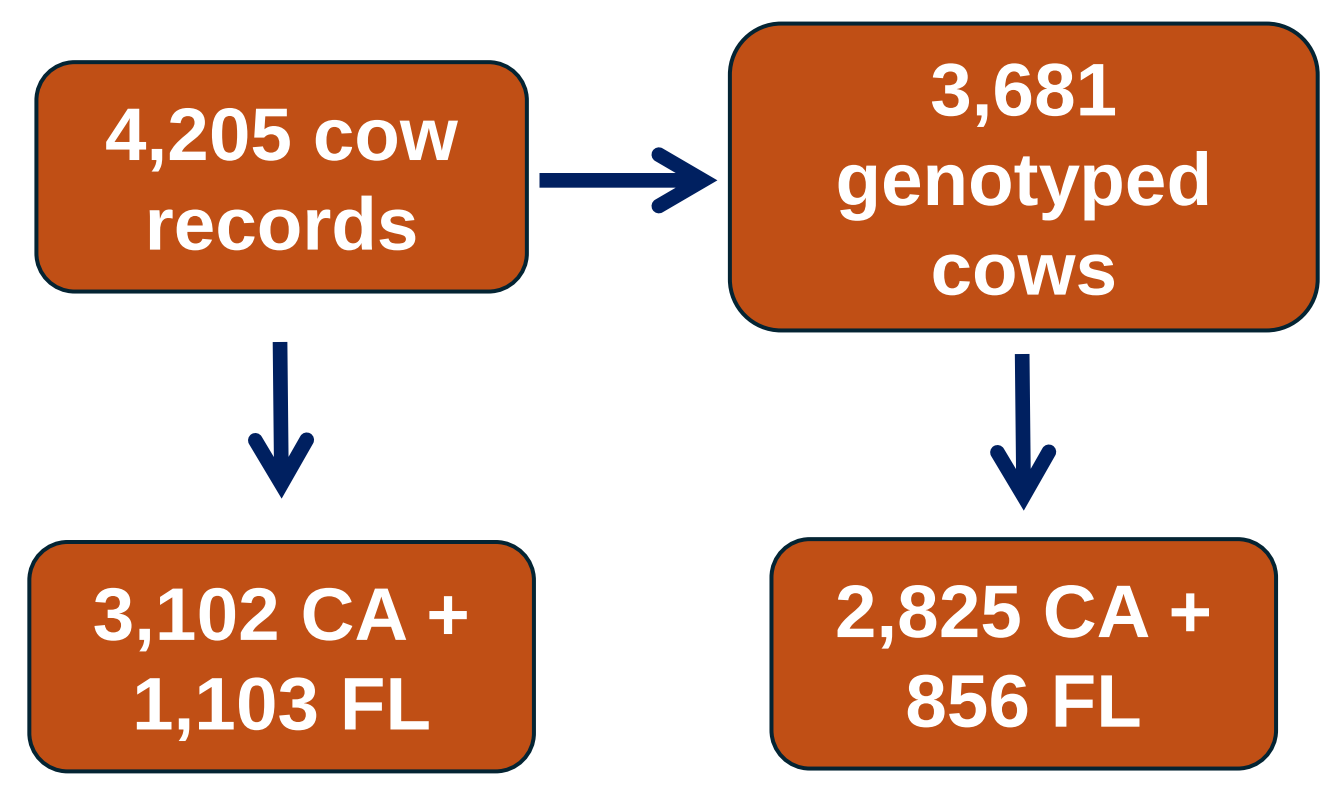
Hypotheses:

- ✓ Genetic and prepartum phenotypic factors are associated with colostrum traits; and
- ✓ Colostrum and milk yield are positively associated.

Objectives:

- ✓ Identify risk factors for colostrum yield and composition;
- ✓ Estimate heritability and genetic correlations of colostrum traits; and
- ✓ Evaluate the relationship between colostrum and milk yield.

MATERIALS AND METHODS



Data from 11 experiments with a total of 38 treatments

Colostrum measurements:

- ✓ Yield recorded at first milking after calving;
- ✓ Analyzed for:
 - Immunoglobulin G;
 - Fat;
 - True protein; and
 - Lactose.
- ✓ Net energy calculated.

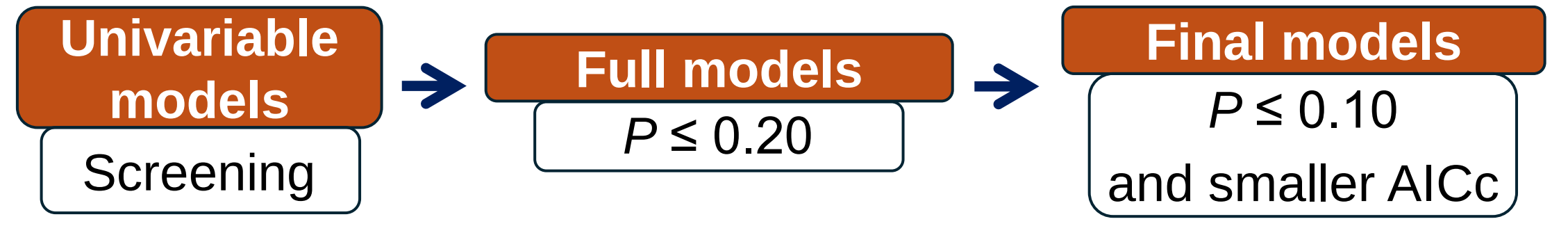
Phenotypic data collected:

- ✓ Calving season;
- ✓ Parity prepartum;
- ✓ Dry period length;
- ✓ Gestation length;
- ✓ Body condition score;
- ✓ Cow body weight (BW);
- ✓ Calving difficulty;
- ✓ Retained placenta
- ✓ Metritis;
- ✓ Calf birth BW;
- ✓ Calf sex;
- ✓ Days in prepartum group; and
- ✓ Prepartum intakes of DM, MP and NE.



Statistical analysis:

- ✓ Mixed-effects models using SAS/STAT (version 9.4) with treatment nested within experiment as a random effect;



- ✓ Multicollinearity was assessed using variance inflation factor (VIF) and variables with VIF > 5 were not included in the same model;

Genetic parameters:

- ✓ Heritability and genetic correlations estimated using **single- and multi-trait GBLUP models**, with variance components obtained by *airemlF90* (BLUPF90 software);
- ✓ $h^2 = \frac{\sigma^2a}{\sigma^2p}$ Where: σ^2a is the additive genetic variance and σ^2p is the phenotypic variance.

RESULTS

- ✓ Colostrum yield was influenced by calving season, parity, calf sex, calf birth BW, gestation length, and days dry;
- ✓ IgG concentration in colostrum was influenced by calving season, parity, and days dry;
- ✓ Prepartum intakes of DM, MP, and NE were not associated with colostrum yield or with IgG concentration in colostrum; and
- ✓ Colostrum yield (kg) was positively associated ($P < 0.0001$) with milk yield (kg/d) during the first 10 weeks postpartum ($\beta = 0.13 \pm 0.03$).

Table 1. Associations of prepartum DM and nutrient intake with colostrum yield (kg) and IgG content (g/L)

Predictor for colostrum yield*	$\beta \pm SE$ (kg/d)	P-value
DM intake, kg/d	-0.11 ± 0.09	0.20
NE intake, Mcal/d	-0.08 ± 0.06	0.14
MP intake, g/d	-0.001 ± 0.0009	0.13
MP balance, g/d	-0.001 ± 0.0006	0.07

Predictor for IgG content*	$\beta \pm SE$ (g/L)	P-value
DM intake, kg/d	0.24 ± 0.55	0.67
NE intake, Mcal/d	0.22 ± 0.34	0.52
MP intake, g/d	0.004 ± 0.006	0.48
MP balance, g/d	0.01 ± 0.006	0.08

*Models retained the fixed effect of nutrient intake and the random effects of treatment nested within experiment and the interaction between nutrient intake and treatment nested with experiment.

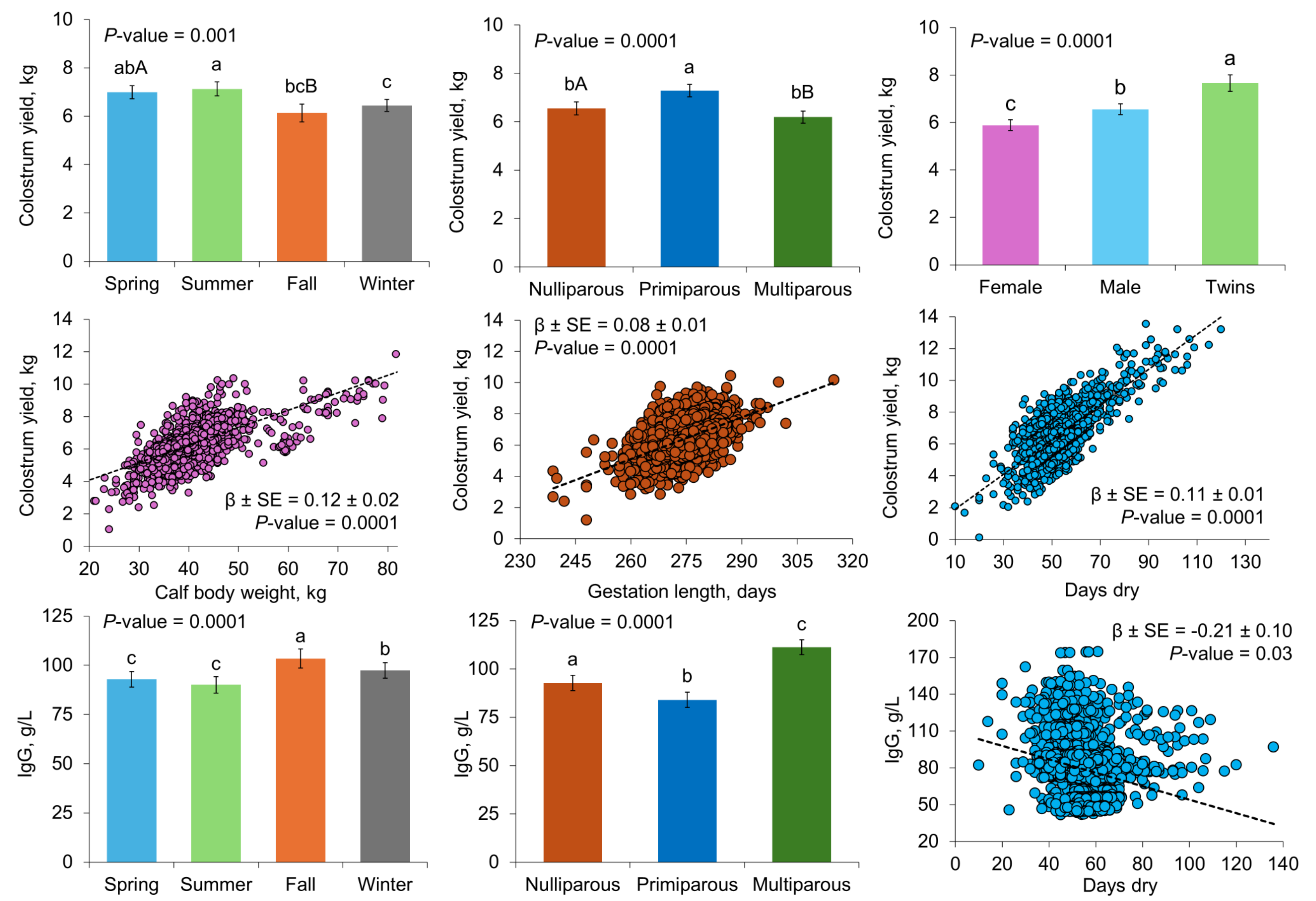


Table 2. Five-trait heritabilities in the diagonal, phenotypic correlations above diagonal, and genetic correlations below diagonal ($\pm SE$)

	Yield (kg)	Fat (%)	Protein (%)	IgG (g/L)	Total IgG (g)
Yield (kg)	0.30 (0.03)	0.19 (0.02)	-0.18 (0.02)	-0.17 (0.02)	0.65 (0.01)
Fat (%)	0.18 (0.07)	0.31 (0.03)	-0.18 (0.02)	-0.05 (0.02)	0.17 (0.02)
Protein (%)	-0.12 (0.07)	-0.02 (0.07)	0.34 (0.03)	0.80 (0.01)	0.36 (0.02)
IgG (g/L)	-0.12 (0.07)	0.05 (0.08)	0.96 (0.01)	0.28 (0.03)	0.50 (0.01)
Total IgG (g)	0.77 (0.07)	0.24 (0.08)	0.50 (0.06)	0.52 (0.06)	0.25 (0.03)

CONCLUSIONS

- ✓ Colostrum yield and composition were associated with both phenotypic and genetic factors.
- ✓ Prepartum intake of DM, MP or NE were not associated with yield or composition of colostrum.
- ✓ The moderate heritability for yield and composition of colostrum suggests that these traits can be improved through genetic selection.